

this value. 2. The plant could make eight different gametes (YRI, YRi, YrI, Yri, yRI, yRi, yrI, and yri). To fit all the possible gametes in a self-pollination, a Punnett square would need 8 rows and 8 columns. It would have spaces for the 64 possible unions of gametes in the offspring. 3. Self-pollination is sexual reproduction because meiosis is involved in forming gametes, which unite during fertilization. As a result, the offspring in self-pollination are genetically different from the parent. (As mentioned in the footnote near the beginning of Concept 14.1, we have simplified the explanation in referring to the single pea plant as a parent. Technically, the gametophytes in the flower are the two "parents.")

Concept Check 14.2

1. $\frac{1}{2}$ homozygous dominant (AA), 0 homozygous recessive (aa), and $\frac{1}{2}$ heterozygous (Aa) 2. $\frac{1}{4}$ BBDD; $\frac{1}{4}$ BbDD; $\frac{1}{4}$ BBdd; $\frac{1}{4}$ Bbdd 3. The genotypes that fulfill this condition are ppyyII, ppyyIi, ppYYii, ppYyii, PpyyII, and ppyyIi. Use the multiplication rule to find the probability of getting each genotype, and then use the addition rule to find the overall probability of meeting the conditions of this problem:

$$\begin{aligned} ppyyII & \frac{1}{2}(\text{probability of } pp) \times \frac{1}{4}(yy) \times \frac{1}{4}(II) = \frac{1}{32} \\ ppyyIi & \frac{1}{2}(pp) \times \frac{1}{4}(yy) \times \frac{1}{2}(Ii) = \frac{1}{16} = \frac{2}{32} \\ ppYYii & \frac{1}{2}(pp) \times \frac{1}{4}(YY) \times \frac{1}{4}(ii) = \frac{1}{32} \\ ppYyii & \frac{1}{2}(pp) \times \frac{1}{2}(Yy) \times \frac{1}{4}(ii) = \frac{1}{16} = \frac{2}{32} \\ PpyyII & \frac{1}{2}(Pp) \times \frac{1}{4}(yy) \times \frac{1}{4}(II) = \frac{1}{32} \\ PpyyIi & \frac{1}{2}(Pp) \times \frac{1}{4}(yy) \times \frac{1}{2}(Ii) = \frac{1}{16} = \frac{2}{32} \\ \hline \text{Fraction predicted to be homozygous recessive} & \frac{8}{32} = \frac{1}{4} \\ \text{for at least two of the three characters} & \end{aligned}$$

Concept Check 14.3

1. Incomplete dominance describes the relationship between two alleles of a single gene, whereas epistasis relates to the genetic relationship between two genes (and the respective alleles of each). 2. Half of the children would be expected to have type A blood and half type B blood. 3. The black and white alleles are incompletely dominant, with heterozygotes being gray in color. A cross between a gray rooster and a black hen should yield approximately equal numbers of gray and black offspring.

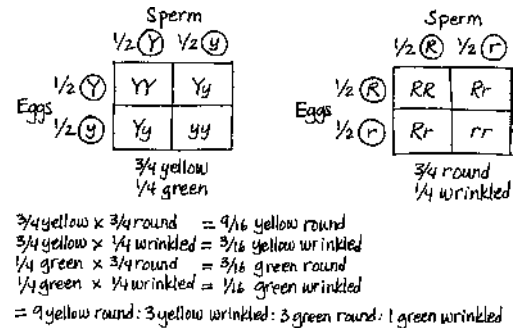
Concept Check 14.4

1. $\frac{1}{9}$ (Since cystic fibrosis is caused by a recessive allele, Lucia and Jared's siblings who have CF must be homozygous recessive. Therefore, each parent must be a carrier of the recessive allele. Since neither Lucia nor Jared has CF and are thus not *cc* (using *C/c* as the alleles for the CF gene), this means they each have a $\frac{2}{3}$ chance of being a carrier. If they are both carriers, there is a $\frac{1}{4}$ chance that they will have a child with CF; $\frac{2}{3} \times \frac{2}{3} \times \frac{1}{4} = \frac{1}{9}$); virtually 0 (Both Lucia and Jared would have to be carriers to produce a child with the disease, unless a very rare mutation (change) occurred in the DNA of cells making eggs or sperm in a noncarrier that resulted in the CF allele.) 2. In normal hemoglobin, the sixth amino acid is glutamic acid (Glu), which is acidic (has a negative charge on its side chain). In sickle-cell hemoglobin, Glu is replaced by valine (Val), which is a nonpolar amino acid, very different from Glu. The primary structure of a protein (its amino acid sequence) ultimately determines the shape of the protein and thus its function. The substitution of Val for Glu enables the hemoglobin molecules to interact with each other and form long fibers, leading to the protein's deficient function and the deformation of the red blood cell. 3. Juanita's genotype is *Dd*. Because the allele for polydactyly (*D*) is dominant to the allele for five digits per appendage (*d*), the trait is expressed in people with either the *DD* or *Dd* genotype. But because Juanita's father does not have polydactyly, his genotype must be *dd*, which means that Juanita inherited a *d* allele from him. Therefore, Juanita, who does have the trait, must be heterozygous. 4. In the monohybrid cross involving flower color, the ratio is 3.15 purple: 1 white, while in the human family in the pedigree, the ratio in the third generation is 1 taster of PTC: 1 nontaster of PTC. The difference is due to the small sample size (two offspring) in the human family. If the second-generation couple in this pedigree were able to have 929 offspring as in the pea plant cross, the ratio would likely be closer to 3:1. (Note that none of the pea plant crosses in Table 14.1 yielded exactly a 3:1 ratio.)

Summary of Key Concepts Questions

14.1 Alternative versions of genes, called alleles, are passed from parent to offspring during sexual reproduction. In a cross between purple- and white-flowered homozygous parents, the F_1 offspring are all heterozygous, each inheriting a purple allele from one parent and a white allele from the other. Because the purple allele is dominant, it determines the phenotype of the F_1 offspring to be purple, and the expression of the recessive white allele is masked. Only in the F_2 generation is it possible for some of the offspring to be homozygous recessive, which causes the white trait to be expressed.

14.2

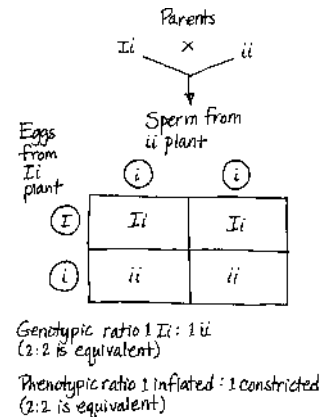


14.3 The ABO blood group is an example of multiple alleles because this single gene has more than two alleles (I^A , I^B , and i). Two of the alleles, I^A and I^B , exhibit codominance since both carbohydrates (A and B) are present when these two alleles exist together in a genotype. I^A and I^B each exhibit complete dominance over the i allele. This situation is not an example of incomplete dominance because each allele affects the phenotype in a distinguishable way, so the result is not intermediate between the two phenotypes. Because this situation involves a single gene, it is not an example of epistasis or polygenic inheritance. 14.4 The chance of the fourth child having cystic fibrosis is $\frac{1}{4}$, as it was for each of the other children, because each birth is an independent event. We already know both parents are carriers, so whether their first three children are carriers or not has no bearing on the probability that their next child will have the disease. The parents' genotypes provide the only relevant information.

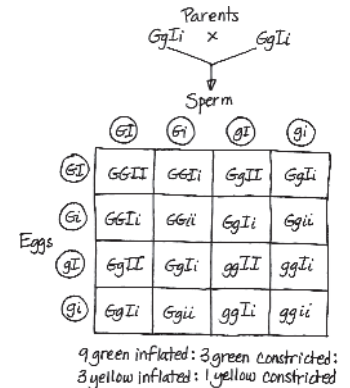
Test Your Understanding

1. A cross of $Ii \times ii$ would yield offspring with a genotypic ratio of 1 Ii : 1 ii (2:2 is an equivalent answer) and a phenotypic ratio of 1 inflated: 1 constricted (2:2 is equivalent).

2. Man $I^A i$; woman $I^B i$; child ii . Genotypes for future children are predicted to be $\frac{1}{4} I^A I^B$, $\frac{1}{4} I^A i$, $\frac{1}{4} I^B i$, $\frac{1}{4} ii$. 3. $\frac{1}{2}$



4. The green pod trait is dominant, so the green pod allele is *G* and the yellow pod allele is *g*; the inflated pod trait is dominant, so the inflated pod allele is *I* and the constricted pod allele is *i*. The cross described is $GgIi \times GgIi$.



5. (a) $\frac{1}{64}$; (b) $\frac{1}{64}$; (c) $\frac{1}{8}$; (d) $\frac{1}{32}$ 6. (a) $\frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} = \frac{27}{64}$; (b) $1 - \frac{27}{64} = \frac{37}{64}$; (c) $\frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} = \frac{1}{64}$; (d) $1 - \frac{63}{64} = \frac{1}{64}$ 7. (a) $\frac{1}{256}$; (b) $\frac{1}{16}$; (c) $\frac{1}{256}$; (d) $\frac{1}{64}$; (e) $\frac{1}{128}$ 8. (a) 1; (b) $\frac{1}{32}$; (c) $\frac{1}{8}$; (d) $\frac{1}{2}$ 9. $\frac{1}{9}$

10. Matings of the original mutant cat with true-breeding noncurl cats will produce both curl and noncurl F_1 offspring if the curl allele is dominant, but only noncurl offspring if the curl allele is recessive. Whether the curl trait is dominant or recessive, you would obtain some true-breeding offspring homozygous for the curl allele from matings between the F_1 cats resulting from the original curl $\times$ noncurl crosses. If dominant, you wouldn't be able to tell true-breeding, homozygous offspring from heterozygotes without further crosses. You know that cats are true-breeding when curl $\times$ curl matings produce only curl offspring. As it turns out, the allele that causes curled ears is dominant. 11. 25%, or $\frac{1}{4}$, will be cross-eyed; all (100%) of the cross-eyed offspring will also be white. 12. The dominant allele *I* is epistatic to the *P/p* locus, and thus the genotypic ratio for the F_1 generation will be 9 $I-P-$ (colorless) : 3 $I-pp$ (colorless) : 3 $iiP-$ (purple) : 1 $iipp$ (red). Overall, the phenotypic ratio is 12 colorless : 3 purple : 1 red. 13. Recessive.